Software Requirements Specification (SRS)

Project: Student Attendance Management System
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Revision history

Version	Date	Author	Change summary	Approval
1.0	01-10-2026	Student team no 1	SRS for Student Attendance Management System	

Approvals

Role	Name	Signature / Email	Date
Course Coordinator	Arpitha k		

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1. Introduction

1.1 Purpose

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1.2 Scope

Covers user authentication, class and student management, attendance marking and correction, attendance history, percentage calculation, low-attendance identification, and automated report generation. It excludes institutional ERP internals and physical biometric hardware beyond defined interfaces.

1.3 Audience

Students, Faculty, Administrators, Developers, QA Engineers, System Integrators, and Assessment Evaluators.

1.4 Definitions

List of acronyms: SAM (Student Attendance Management), UI, API, RBAC, CSV, HTTPS, TLS, RTM.

2. Overall description

2.1 Product perspective

The system is a web-based application used by students, faculty, and administrators. It stores student, course, class, enrollment, attendance, and audit data and provides attendance summaries and automated reports through application APIs.

2.2 Major product functions (detailed)

- Authenticate students, faculty, and administrators
- Manage students, faculty, courses, classes, and enrollments
- Mark Present, Absent, or Late attendance
- Prevent duplicate attendance records
- Correct attendance with audit information
- View attendance history and percentages
- Identify low-attendance students
- Generate and export attendance reports
- Provide attendance dashboards and audit records

2.3 User roles and characteristics (expanded)

- Student: views personal attendance history and percentages; cannot modify attendance.

- Faculty: marks and corrects attendance for assigned classes and generates reports.
- Administrator: manages accounts, classes, courses, enrollments, and system-level reports.

2.4 Operating environment

Web browser on desktop or mobile devices, application server, relational database server, and secure HTTPS network connection.

2.5 Constraints

Attendance access is controlled by user roles and class assignments. Attendance percentage and low-attendance thresholds are configurable. The system depends on a relational database and secure application-server deployment.

3. External interface requirements

3.1 User interfaces

Primary UI: responsive web interface with separate dashboards for students, faculty, and administrators. Forms shall provide clear validation messages and accessible controls.

3.2 Hardware interfaces

No specialized hardware is required. The system may be accessed through standard computers, tablets, or smartphones using a browser.

3.3 Software interfaces

- Authentication/API layer for login and role validation.
- Relational database interface for student, class, enrollment, attendance, and audit data.
- Report/export interface for CSV generation.

3.4 Communications

- HTTPS/TLS shall be enforced for application traffic.
- API requests shall use authenticated sessions/tokens.
- Database connections shall use secured server-side credentials.

4. System features (detailed)

Each requirement below includes acceptance criteria and a reference test case. IDs follow SAM-F-###.

4.1 Authentication & Authorization

Description: Authenticate registered students, faculty, and administrators and enforce role-based access.

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Req ID	Requirement	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
SAM-F-001	The system shall authenticate users using a registered username and password.	Functional	High	Students / Faculty / Admin	AC-SAM-F-001: Valid credentials create an authenticated session. Test: TC-AUTH-01	Requires user account database
SAM-F-002	The system shall enforce role-based access for students, faculty, and administrators.	Functional	High	Security / Admin	AC-SAM-F-002: Users can access only permitted functions. Test: TC-AUTH-02	Requires RBAC configuration
SAM-F-003	The system shall allow faculty to select a class and attendance date.	Functional	High	Faculty	AC-SAM-F-003: Faculty can select an assigned class and valid date. Test: TC-ATT-01	Requires class assignment data
SAM-F-004	The system shall display the enrolled student roster for the selected class.	Functional	High	Faculty	AC-SAM-F-004: Correct enrolled students are displayed. Test: TC-ATT-01	Requires roster data

SAM-F-005	The system shall allow faculty to mark each enrolled student as Present, Absent, or Late.	Functional	High	Faculty	AC-SAM-F-005: Each roster student receives one valid status. Test: TC-ATT-02	Attendance policy is configurable
SAM-F-006	The system shall prevent duplicate attendance records for the same student, class, and date.	Functional	High	Data Integrity	AC-SAM-F-006: Duplicate submission is rejected or updates the existing record. Test: TC-ATT-03	Unique attendance constraint required
SAM-F-007	The system shall allow authorized faculty to edit an attendance record with an audit timestamp.	Functional	High	Faculty / Admin	AC-SAM-F-007: Authorized correction is saved and timestamped. Test: TC-ATT-04	Audit fields required
SAM-F-008	The system shall allow students to view their attendance history by course and date.	Functional	Medium	Students	AC-SAM-F-008: Student sees only their own records. Test: TC-STU-01	Requires authenticated student

SAM-F-009	The system shall calculate attendance percentage for each student and course.	Functional	High	Faculty / Students	AC-SAM-F-009: Percentage matches attendance records and configured policy. Test: TC-REP-01	Calculation must use stored records

4.2 Attendance Capture

Description: Record attendance for enrolled students for a selected class and date while preventing duplicates and supporting authorized corrections.

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Req ID	Requirement (shall...)	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
SAM-F-010	The system shall identify students whose attendance falls below the configured threshold.	Functional	High	Faculty / Admin	AC-SAM-F-010: Students below threshold are flagged. Test: TC-REP-02	Threshold is configurable
SAM-F-011	The system shall generate attendance reports for a selected class, course, and date range.	Functional	High	Faculty / Admin	AC-SAM-F-011: Report contains correct students, counts, and percentages. Test: TC-REP-03	Requires report service

SAM-F-012	The system shall support filtering reports by student, class, course, and date range.	Functional	Medium	Faculty / Admin	AC-SAM-F-012: Filters return only matching records. Test: TC-REP-04	Depends on indexed fields
SAM-F-013	The system shall export generated attendance reports in CSV format.	Functional	Medium	Faculty / Admin	AC-SAM-F-013: Export contains the displayed report data. Test: TC-REP-05	File generation required
SAM-F-014	The system shall allow administrators to create, update, and deactivate student and faculty accounts.	Functional	High	Administrator	AC-SAM-F-014: Account changes are persisted and reflected in login. Test: TC-ADM-01	Admin-only operation
SAM-F-015	The system shall allow administrators to create and manage classes, courses, and student enrollments.	Functional	High	Administrator	AC-SAM-F-015: Enrollment changes appear in the class roster. Test: TC-ADM-02	Referential integrity required

SAM-F-016	The system shall maintain an audit trail for attendance corrections and administrative changes.	Functional	High	Security / Admin	AC-SAM-F-016: Actor, action, target, and timestamp are recorded. Test: TC-AUD-01	Audit data protected from ordinary users
SAM-F-017	The system shall provide a dashboard showing attendance summaries and report-generation status.	Functional	Medium	Faculty / Admin	AC-SAM-F-017: Dashboard displays current summary values. Test: TC-DASH-01	Depends on report calculations

4.3 Attendance History & Calculation

Requirements: The system shall provide attendance history, percentage calculations, and low-attendance identification. IDs: SAM-F-008..SAM-F-010.

4.4 Reports & Analytics

Requirements: The system shall generate filtered attendance reports and support CSV export. IDs: SAM-F-011..SAM-F-013.

4.5 Administration & Audit

Requirements: Provide administrator functions for accounts, classes, enrollments, audit records, and dashboard summaries. IDs: SAM-F-014..SAM-F-017.

5. Non-functional requirements (detailed)

NFRs below are measurable and tied to test plans. IDs SAM-NF-###.

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement

SAM-NF-001	Common attendance and report operations shall respond within 3 seconds for 95% of requests under normal load.	Performance	High	95th percentile ≤ 3s. Test: TC-PERF-01
SAM-NF-002	The system shall provide at least 99.5% monthly availability excluding scheduled maintenance.	Reliability	High	Uptime report ≥99.5% per month. Test: TC-OPS-01
SAM-NF-003	All authenticated communication between browser and server shall use HTTPS/TLS.	Security	High	Security scan shows no plaintext application traffic. Test: TC-SEC-01
SAM-NF-004	The system shall preserve attendance data using transactional database operations and regular backups.	Reliability/Data Integrity	High	Recovery test restores a valid backup without inconsistent attendance records. Test: TC-REC-01
SAM-NF-005	The user interface shall be usable on common desktop and mobile browser screen sizes.	Usability	Medium	Responsive UI review passes defined viewport checks. Test: TC-UX-01

SAM-NF-006	The system shall retain audit records for at least one academic year or the configured institutional retention period.	Audit/Data Retention	Medium	Audit records remain retrievable for the configured period. Test: TC-AUD-02

5.1. Security

5.1.1 Security Objectives

Security objectives:
- Protect attendance and account data from unauthorized access or modification.
- Provide accountability through authenticated access and audit trails.

5.1.2 Security Requirements

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case ref
SAM-SR-001	The system shall enforce HTTPS/ TLS for all authenticated network communication.	Security	High	TC-SEC-01
SAM-SR-002	The system shall store passwords only as salted, one-way password hashes and shall never log plaintext passwords.	Security	High	TC-SEC-02
SAM-SR-003	The system shall enforce role-based authorization on every protected API operation.	Security	High	TC-SEC-03
SAM-SR-004	The system shall record security-relevant login failures, attendance corrections, and administrative changes in an audit log.	Security	High	TC-AUD-01
SAM-SR-005	The system shall protect against unauthorized access to another student’s attendance data through server-side authorization checks.	Security	High	TC-SEC-04

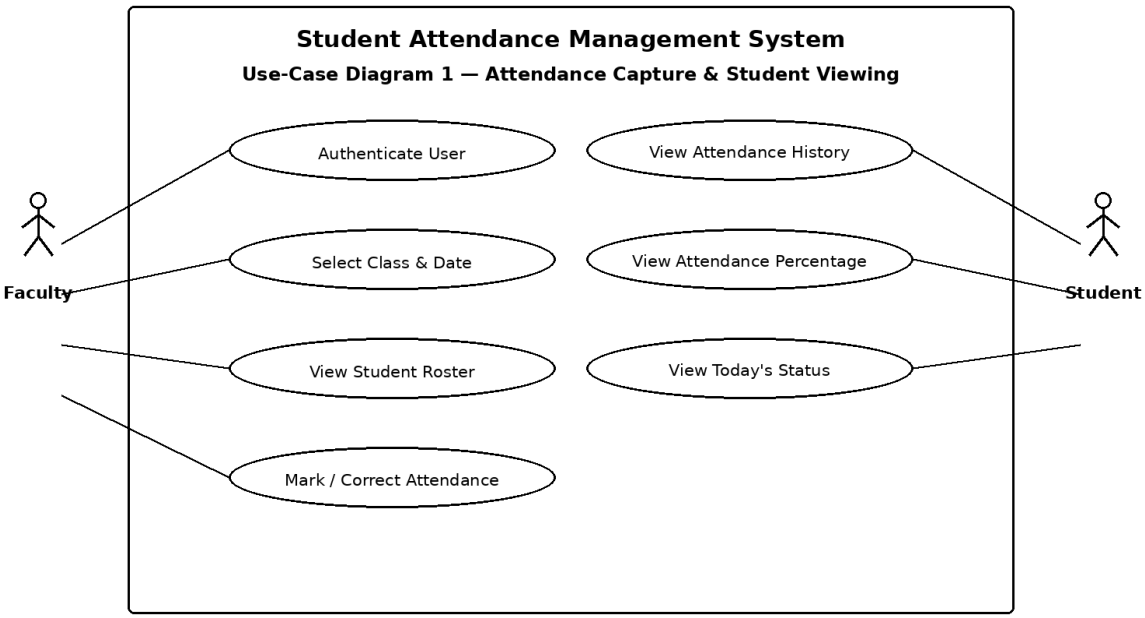
6. Quality attributes & Acceptance tests

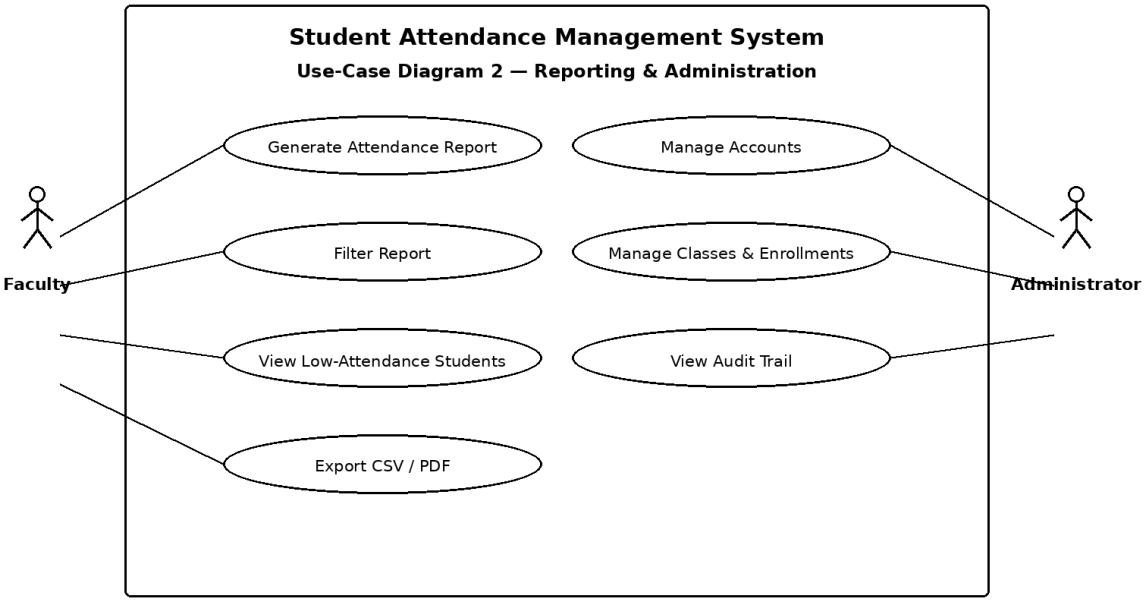
- Exit criteria for acceptance: All high-priority functional requirements implemented and verified, no critical NFR failures, and RTM shows all planned test cases mapped and executed.
- Acceptance threshold: attendance calculations and reports shall match the underlying attendance records.
- Acceptance test suites: Authentication, Attendance Marking, Attendance Correction, Reporting, Performance, Security, Recovery, and Usability tests.

7. System models and diagrams

7.1 UML Use-Case diagram

Two UML use-case diagrams are provided: (1) Attendance capture and student attendance viewing; (2) Reporting and administration.




Student Attendance Management System
Use-Case Diagram 2 — Reporting & Administration
Generate Attendance Report
Filter Report
View Low-Attendance Students
Export CSV / PDF
Manage Accounts
Manage Classes & Enrollments
View Audit Trail
Faculty
Administrator

8. Requirements Traceability Matrix (RTM)

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
SAM-F-001	Authenticate users	4.1/Auth	Auth/RBAC	TC-AUTH-01	N	Requires registered user accounts.
SAM-F-002	Enforce RBAC	4.1/Auth	Auth/RBAC	TC-AUTH-02	N	Access permissions must match each user's role.
SAM-F-003	Select class/date	4.2/Attendance	Attendance	TC-ATT-01	N	Depends on class assignments and valid attendance dates.
SAM-F-004	Display roster	4.2/Attendance	Attendance	TC-ATT-01	N	Roster must reflect current student enrollments.
SAM-F-005	Mark attendance	4.2/Attendance	Attendance	TC-ATT-02	N	Allowed statuses follow the configured attendance policy.
SAM-F-006	Prevent duplicates	4.2/Attendance	Attendance	TC-ATT-03	N	Requires a unique student/class/date attendance record.
SAM-F-007	Correct attendance	4.2/Attendance	Attendance	TC-ATT-04	N	Corrections must be restricted to authorized users and audited.

SAM-F-008	View attendance history	4.3/ History	Student View	TC-STU-01	N	Students may view only their own attendance records.
SAM-F-009	Calculate percentage	4.3/ History	Report	TC-REP-01	N	Calculation depends on accurate records and configured rules.
SAM-F-010	Flag low attendance	4.4/ Reports	Report	TC-REP-02	N	Uses the configured low-attendance threshold.
SAM-F-011	Generate reports	4.4/ Reports	Report	TC-REP-03	N	Report totals must match the underlying attendance records.
SAM-F-012	Filter reports	4.4/ Reports	Report	TC-REP-04	N	Filters must be applied consistently to report results.
SAM-F-013	Export CSV	4.5/Admin	Report	TC-REP-05	N	Exported CSV data must match the displayed report.
SAM-F-014	Manage accounts	4.5/Admin	Admin	TC-ADM-01	N	Account-management actions are administrator-only.
SAM-F-015	Manage classes/ enrollments	4.5/Admin	Admin	TC-ADM-02	N	Enrollment changes must be reflected in class rosters.
SAM-F-016	Audit changes	4.5/Admin	Audit	TC-AUD-01	N	Audit entries should record actor, action, target, and timestamp.
SAM-F-017	Dashboard summaries	4.5/Admin	Dashboard	TC-DASH-01	N	Dashboard values depend on current attendance and report data.

SAM-NF-001	Response time	4.5/Admin	Platform	TC-PERF-01	N	Performance target applies under normal operating load.
SAM-NF-002	Availability	5/NFR	Platform	TC-OPS-01	N	Availability excludes scheduled maintenance.
SAM-NF-003	HTTPS/ TLS	5/NFR	Security	TC-SEC-01	N	Requires HTTPS/TLS configuration across application traffic.
SAM-NF-004	Backup/ recovery	5/NFR	Recovery	TC-REC-01	N	Depends on scheduled backups and a verified recovery process.
SAM-NF-005	Responsive UI	5/NFR	UI	TC-UX-01	N	Usability must be checked on supported desktop and mobile sizes.
SAM-NF-006	Audit retention	5/NFR	Audit	TC-AUD-02	N	Retention period follows the configured institutional policy.
SAM-SR-001	HTTPS/ TLS	5/NFR	Security	TC-SEC-01	N	Applies to all authenticated client-server communication.
SAM-SR-002	Password hashing	5.1.2/Sec	Security	TC-SEC-02	N	Passwords must use salted one-way hashes; plaintext must not be logged.
SAM-SR-003	API authorization	5.1.2/Sec	Security	TC-SEC-03	N	Authorization checks are required on every protected API operation.

SAM-SR-004	Security audit logging	5.1.2/Sec	Audit	TC-AUD-01	N	Security-relevant failures and changes must be logged without exposing secrets.
SAM-SR-005	Student data isolation	5.1.2/Sec	Security	TC-SEC-04	N	Server-side checks must prevent access to another student's records.